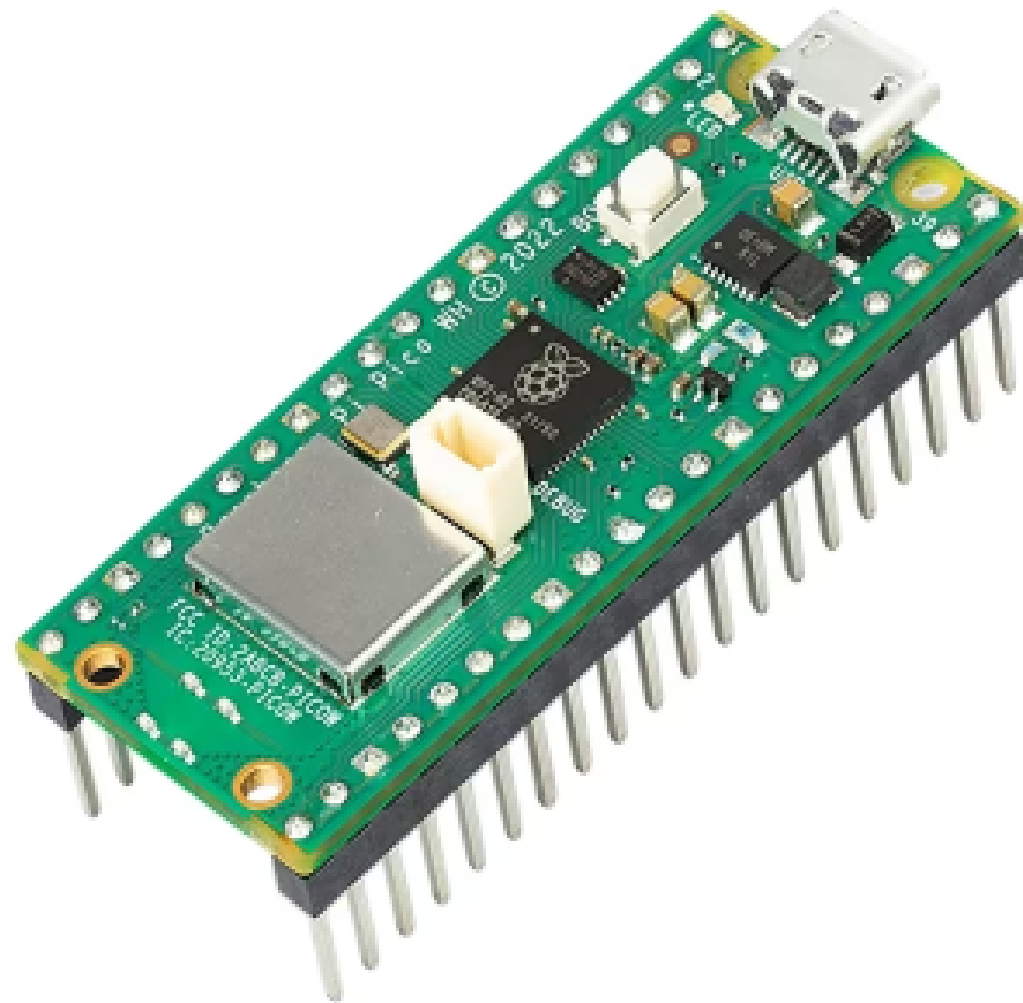
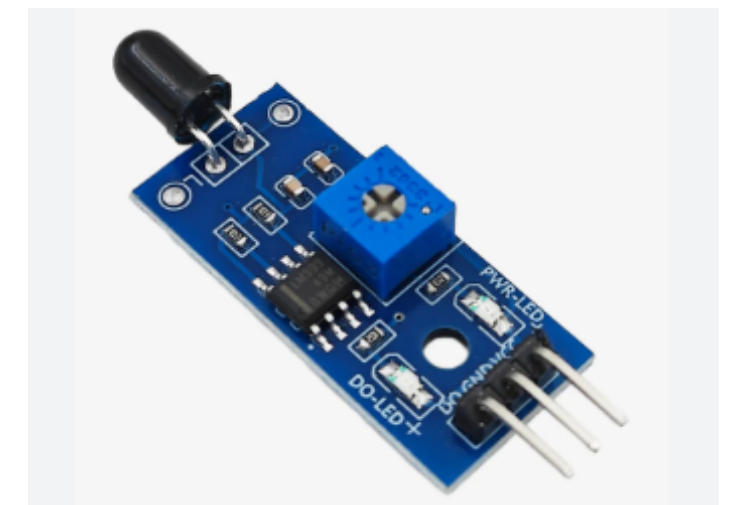


RAKSHAK SMART SAFETY NETWORK



Team Members

- Jui Adke - 1
- Riya Balani - 6
- Riddhi Jadhav - 23
- Ayush Koli - 31



Problem Statement:

- Traditional safety systems rely on single-sensor triggers, making them prone to false alarms or missed detections.
- They cannot analyze multiple hazards (gas leaks, fire, temperature rise, intrusion) simultaneously in real time.
- It has a special innovation that it senses integration of human awareness—by tracking pulse and presence, the system can determine whether someone is actually in danger and adjust its response accordingly.
- There is a need for an intelligent, multi-sensor safety system that can make decisions, reduce false alerts, and provide real-time, accurate responses.

Project title: Rakshak Smart Safety Network

Project Description:

Multi-Sensor Hazard Detection:

The system uses a combination of sensors (MQ gas sensor, flame sensor, temperature sensor, and PIR/ultrasonic sensor) to continuously monitor for dangers such as gas leaks, fire, overheating, and intrusion.

Human-Aware Safety Feature:

By incorporating presence and pulse detection, the system can determine whether a person is at risk and adjust its response accordingly, making it more effective than traditional safety systems.

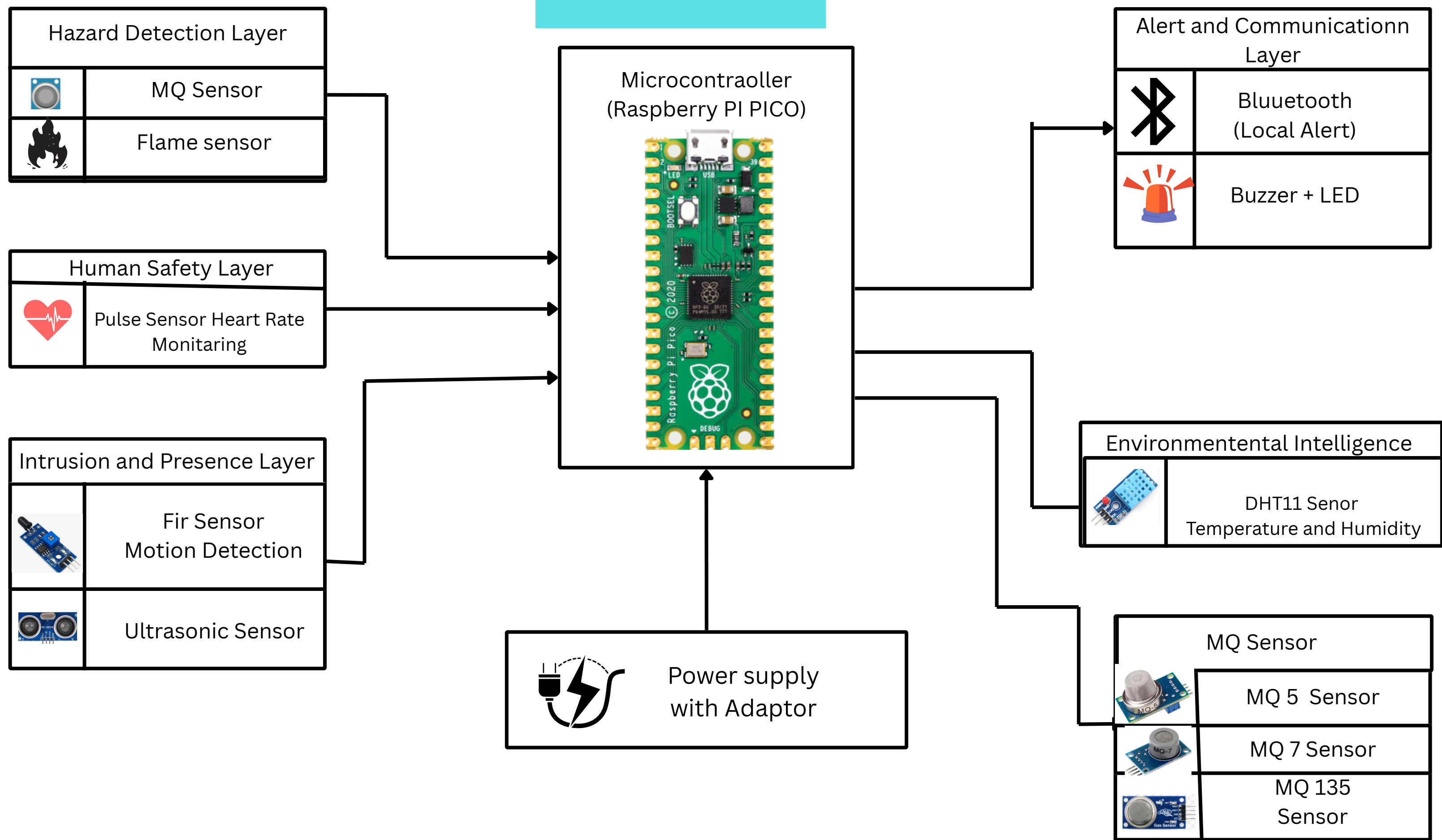
Intelligent Decision-Making:

Instead of reacting to a single trigger, the system analyzes data from multiple sensors to verify threats and reduce false alarms, ensuring more accurate and reliable responses.

Real-Time Alerts and Response:

Using Bluetooth communication, the system sends instant alerts to a mobile device and can take appropriate actions like activating warnings or safety measures based on the severity of the situation.

Block diagram

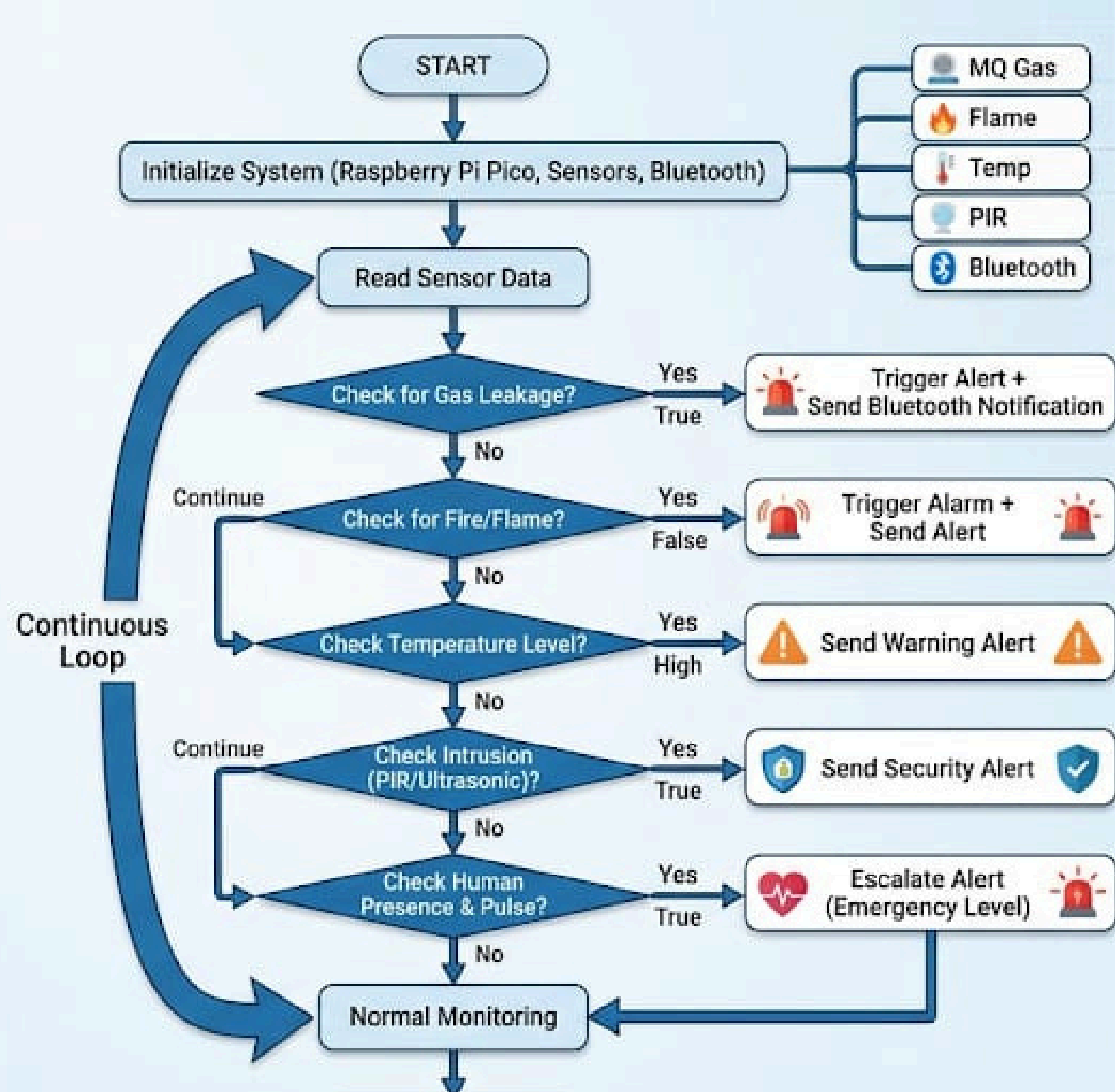


Component	Quantity	Purpose
[Raspberry Pi Pico]	2	[Main controller]
[DHT 11]	1	[Temprature sensor]
[HC-05 Bluetooth module]	1	[Bluetooth connection with external device]
[PPG PULSE sensor]	1	[detects pulse rate]
[3.7v battery]	1	[charge the system]
[MQ 4]	1	[Detects methane]
[MQ 5]	1	[Detects LPG/hydrogen/natural gas]
[MQ 7]	1	[Detects carbon monoxide]
[MQ 135]	1	[Detecs hazard gases]
[Buzzer]	1	[alert system]
[LM3 93]	1	[Senses flame]

Component	Quantity	Total Price
[Raspberry Pi Pico]	2	₹770
[DHT 11]	1	₹75
[HC-05 Bluetooth module]	1	₹265
[PPG PULSE sensor]	1	₹310
[3.7v battery]	1	₹140
[MQ 4]	1	₹120
[MQ 5]	1	₹120
[MQ 7]	1	₹120
[MQ 135]	1	₹120
[Buzzer]	1	₹20
[LM3 93]	1	₹75
		Total- ₹2,275

Flow Chart

Smart Disaster Prevention Hub



Conclusion

The project successfully demonstrates a multi-sensor smart safety system capable of detecting various hazards such as gas leaks, fire, temperature rise, and intrusion

Real-time communication through Bluetooth ensures that users receive instant alerts and can take timely action

It improves reliability by using combined sensor inputs and decision-based logic, reducing false alarms compared to traditional systems.

The system is cost-effective, scalable, and adaptable, making it suitable for homes, industries, and public safety applications

The integration of human awareness (pulse and presence detection) enhances the system's ability to respond more effectively in real-life situations

Overall, the project highlights how embedded systems and sensor integration can be used to build intelligent, responsive, and efficient safety solutions